

Astroviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**



**World Health
Organization**

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Suggested citation. Astroviruses: background document for the WHO guidelines for drinking-water quality and the WHO guidelines on sanitation and health. Geneva: World Health Organization; 2025 (Drinking-water and sanitation related pathogens series). <https://doi.org/10.2471/B09237>. Licence: [CC BY-NC-SA 3.0 IGO](https://creativecommons.org/licenses/by-nc-sa/3.0/igo).

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Contents

Highlights.....	1
1 General description.....	2
2 Human health effects	2
3 Epidemiology	2
4 Transmission	3
5 Detection methods	3
6 Source and occurrence	4
6.1 Faecal pollution sources.....	4
6.2 Surface water	4
6.3 Groundwater	4
6.4 Drinking-water.....	4
6.5 Soil and crops.....	5
6.6 Surfaces.....	5
7 Significance of sanitation.....	5
8 Significance in drinking-water	5
9 Prevention and management	6
9.1 Sanitation systems.....	6
9.2 Drinking-water supplies.....	6
10 ICD-11 classification.....	7
Selected references	8
Annex. Approach to content development and declarations of interest	10

Acknowledgements

This background document on Astroviruses supports the World Health Organization (WHO) Guidelines for drinking-water quality (GDWQ) and Guidelines on sanitation and health.

This document was prepared by S. McFadyen, Health Canada, Canada, and W. Robertson, Canada, independent consultant, under the coordination of WHO. The coordinators were S. Boisson, J. De France, J. Strandberg, WHO, Switzerland. Strategic direction was provided by B. Gordon, WHO. The document received further inputs from the following reviewers and other contributors: O. Cumming, London School of Hygiene & Tropical Medicine, United Kingdom of Great Britain and Northern Ireland; E. Holmes, University of North Carolina, United States of America; D. Medeiros, Health Canada, Canada; L. Mendoza, Stanford University, United States of America; T. Miura, National Institute of Public Health, Japan; S. Petterson, Griffith University, Australia; K. Rotert, US Environmental Protection Agency, United States of America.

The work of the following experts was crucial in overseeing the development of the series of microbial background documents: N.J. Ashbolt, University of South Australia, Australia; J. Brown, University of North Carolina at Chapel Hill, United States of America; E. Calderon, Buenos Aires University, Argentina; D. Cunliffe, South Australian Department of Health, Australia; L. D'Anglada US Environmental Protection Agency, United States of America; A.M. de Roda Husman, National Institute of Public Health and the Environment (RIVM), Netherlands; P.R. Hunter, University of East Anglia, United Kingdom; S. McFadyen, Health Canada, Canada; G. Medema, KWR Watercycle Research Institute and Kiwa, Netherlands; S. Petterson, Water & Health Pty Ltd and Griffith University, Australia; J. Rose, Michigan State University, United States of America; P. Smeets, KWR Watercycle Research, Netherlands; M. Sobsey University of North Carolina at Chapel Hill, United States of America.

WHO gratefully acknowledges the financial support provided by the Agence Française de Développement, France; the Bill and Melinda Gates Foundation, United States of America; the Directorate General for International Cooperation, Netherlands; the Foreign, Commonwealth and Development Office, United Kingdom; the Ministry of Health, Labour and Welfare, Japan; the United States Environmental Protection Agency, United States of America.

All technical and funding inputs from institutions and individuals from the United States of America preceded 20 January 2025.

Highlights

- There is strong evidence linking transmission of astroviruses with inadequate sanitation and poor hygiene. However, the strength of evidence that astroviruses are transmitted through drinking-water is considered weak. Astroviruses are, therefore, of low health significance in drinking-water.
- Humans are the only known source for strains of astrovirus that cause human disease (the so-called classic human astroviruses, HAsVs), but the genetic similarities between some human and animal strains make zoonotic transmission a possibility.
- Transmission is primarily by the faecal–oral route, with person-to-person spread as the most common route of exposure. Contaminated food and fomites have also been identified as exposure sources. Transmission through ingestion of drinking-water is plausible but has not been confirmed.
- Infections may be asymptomatic or cause normally mild and self-limiting symptoms, including diarrhoea with fever, abdominal pain and dehydration.
- HAsVs affect children worldwide, with limited reports of disease in normally healthy adults. Immunocompromised individuals and the elderly are at high risk. In low-income settings, astrovirus is one of the leading aetiological agents for diarrhoea in infants. In temperate climates, infections are more common in winter, while in tropical areas, infections are more common in rainy seasons.
- The ID₅₀ (50% infective dose) for HAsVs has not been determined, but similar to other enteric viruses, the relative infectivity is expected to be high.
- Human astroviruses can be detected in environmental samples using culture and molecular-based techniques. However, there are no standard methods for the detection of astroviruses in water.
- HAsVs are commonly present in raw sewage, treated wastewater and surface waters and have been detected in groundwater and drinking-water. HAsVs can persist for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including possible risks from astroviruses.
- Reduction of HAsVs by wastewater and drinking-water treatment varies depending on the process. Although astroviruses are moderately resistant to disinfection with chlorine, it can be effective provided adequate doses and contact times are applied.
- Due to the smaller size of astroviruses relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their longer environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAsVs in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAsVs.

1 General description

Human and animal strains of astroviruses are classified in the family *Astroviridae* and further divided into two genera: *Mamastrovirus*, which comprise the human and animal astroviruses, and *Avastrovirus*, which comprise the avian astroviruses (ICTV, 2018). Astroviruses have single-stranded, RNA genomes in a non-enveloped, icosahedral capsid with a diameter of approximately 28 nm to 35 nm. In a proportion of the particles, a distinct star-shaped surface structure can be seen by electron microscopy (Clark & McKendrick, 2004).

There are eight serotypes of the so-called classic human astroviruses (HAstV-1 to HAstV-8), with HAstV-1 being the most frequently isolated

There are eight serotypes of the so-called classic human astroviruses (HAstV-1 to HAstV-8), with HAstV-1 being the most frequently isolated (Moser & Schultz-Cherry, 2005; Lopman et al., 2014; Bosch et al., 2014). The number of astrovirus strains associated with human infections has increased in recent years, with two novel groups of highly divergent astroviruses, named MLB (Melbourne) and VA/HMO (Virginia/Human-Mink-Ovine-like), being identified in 2008 and 2009 (Bosch et al., 2014; Vu et al., 2017).

Although there are no known animal reservoirs for HAstVs (Lopman et al., 2014), zoonotic transmission may be possible due to the genetic similarities between some human and animal strains (Vu et al., 2017).

2 Human health effects

HAstVs cause short-term, self-limiting diarrhoea with fever, abdominal pain, and dehydration; vomiting presents less often than in infections with other enteric viruses (Clark & McKendrick, 2004; Moser & Shultz-Cherry, 2005). Symptoms are generally mild in severity, in contrast with diarrhoea associated with norovirus or rotavirus. Infections usually resolve without treatment within 2 to 3 days of symptoms appearing (Clark & McKendrick, 2004; Moser & Shultz-Cherry, 2005). Asymptomatic infection also occurs; rates of approximately 4% have been reported in children but have reached greater than 20% in some studies (Vu et al., 2017).

Although rare, astrovirus infection can cause severe central nervous system (CNS) diseases, including encephalitis, meningitis, and meningoencephalomyelitis, with high mortality rates in severely immunocompromised human hosts (Johnson et al., 2017; Vu et al., 2017). Since HAstVs are transmitted by the faecal–oral route, it has been suggested that invasion of the CNS occurs via the enteric tract as a consequence of immunodeficiency (Reuter et al., 2018).

3 Epidemiology

The global burden of astroviruses is difficult to ascertain; it is believed that many cases are unreported. HAstVs affect children worldwide, with limited reports of disease in normally healthy adults. However, immunocompromised individuals and the elderly are at high risk (Bosh et al., 2014). A recent review estimated the prevalence among children with gastroenteritis to be approximately 5% (Vu et al., 2017). Astroviruses have been identified as a

leading aetiological agent for diarrhoea among children under 2 years of age in a study across eight low- and middle-income countries (Platts-Mills et al., 2015, Olortegui et al., 2018).

Geographically, astrovirus infections and the serotypes causing infection are widespread (Bosch et al., 2014). Serotype 1 (HAstV-1) is the most common, but other serotypes are also widely distributed, some with geographical differences. For example, in the Republic of Korea, China and Japan, HAstV-1 accounted for over 70% of all HAstV infections analysed, while in Mexico, the prevalence of HAstV-1 was low (10%) compared to that of HAstV-2 (42%) and HAstV-4 (23%) (Jeong et al., 2012). Seasonally, in temperate areas, HAstVs infections are higher in winter, while in tropical areas HAstVs infections peak in the rainy season (Bosch et al., 2014).

4 Transmission

HAstVs are transmitted by the faecal–oral route. Direct person-to-person spread is considered the most common route of transmission. Clusters of cases are seen in families, child-care centres, paediatric wards, homes for the elderly and military establishments. HAstVs also persist on fomites at levels high enough to present a risk to vulnerable populations (Vu et al., 2017). Transmission via ingestion of contaminated food has been documented, including links to bivalve molluscs, fresh produce and prepared foods (Oishi et al., 1994; Le Guyader et al., 2008; Bosch et al., 2014; Yates, 2014). Unsafe food handling leads to contamination of ready-to-eat foods that are consumed with little, if any, cooking (Vu et al., 2017). Asymptomatic carriage of astroviruses is not well-characterized but shedding from asymptomatic food workers may contribute to transmission (Bosch et al., 2014). The magnitude of the health risk from waterborne exposures to HAstVs is largely unknown but probably low based on the available evidence of health effects (Gofti-Laroche et al., 2003; Hwang et al., 2015).

The relative infectivity of HAstVs is unknown, but given their similarities to other enteric viruses, it is probably high.¹ The incubation period for HAstVs is between 1 and 5 days, after which infected individuals generally excrete large numbers of HAstVs in faeces (about 110 to 1000 genome copies/g of stool) (Moser & Shultz-Cherry, 2005; Bosch et al., 2014) for up to 2 weeks after symptoms end (Moser & Shultz-Cherry, 2005; Bosch et al., 2014). Given that HAstV infections are seen principally in younger children, immunity probably lasts for many years, particularly against the prevalent classic HAstVs. Yet, with the genetic and antigenic diversity of HAstV serotypes, reinfection can occur (Lopman et al., 2014).

5 Detection methods

Culture-based methods are available for isolation of classic HAstVs from clinical and environmental samples, and molecular-based methods are available for classic and novel human HAstVs (Bosch et al., 2014; da Silva et al., 2016; Corpuz et al., 2020). Although a consensus set of molecular-based primers for HAstVs has been published (da Silva et al., 2016) and next-

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

generation sequencing has been used to detect HAsTVs (Hata et al., 2018), there are no standard methods available for their detection in water. Furthermore, distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods may make it possible to improve the detection of infectious enteroviruses

6 Source and occurrence

6.1 Faecal pollution sources

In studies of raw sewage worldwide, HAsTVs were detected in 29% to 100% of samples in concentrations up to 4.3×10^7 genome copies (GC)/L (da Silva et al., 2016). In studies of secondary treated but undisinfected wastewater, HAsTVs were detected in up to 100% of samples with average concentrations of 10^3 to 10^4 GC/L (da Silva et al., 2016).

6.2 Surface water

Human astroviruses have been detected in rivers, lakes and streams affected by sewage (up to 60% of samples) in concentrations ranging from 2.8×10^1 to 1×10^3 GC/L (da Silva et al., 2016). Using next-generation sequencing, classic, MLB and VA HAsTV strains were detected in 57%, 29% and 36% of samples, respectively, from a river in Nepal (Hata et al., 2018). Shi et al. (2021) detected HAsTV in 83% of samples from two rivers in China. The infectivity of astroviruses was reduced by 4 log₁₀ units after 30 days in surface water at 20 °C (Espinosa et al., 2008).

6.3 Groundwater

Few studies have examined the presence of HAsTVs in groundwater. Classic and MLB strains were detected by next-generation sequencing in 5% and 3%, respectively, of 37 groundwater samples in Nepal (Hata et al., 2018).

6.4 Drinking-water

Human astroviruses have been detected in drinking-water samples in several studies. One study in Slovenia detected HAsTVs by molecular-based methods in 1 of 17 samples from public water supplies described as pretreated and disinfected (Steyer et al., 2011). Human astroviruses were also detected by molecular-based methods in 8.3% of tap water samples (with no information on water treatment provided) from a hospital in China (Shi et al., 2021), 2% (1 of 44) of chlorinated tap water samples with low levels of residual chlorine (<0.02 mg/L) in France (Gofti-Laroche et al., 2003); in 8% (1/12) of filtered and disinfected drinking-water samples in Egypt (Morsy El-Senousy et al., 2007); and at the ends of a distribution system in a public water supply in Ghana, suggesting intrusion of contaminated water (Dongdem et al., 2011). In experiments where astroviruses were added to undisinfected tap water from a groundwater source, 2 log₁₀ and 3.3 log₁₀ reductions in infectivity were measured at 4 °C after 60 days and 90 days, respectively (Abad et al., 1997). At 20 °C, the reduction was greater than 2 log₁₀ after 30 days, 3.3 log₁₀ at 60 days and 5 log₁₀ at 90 days (Abad et al., 1997). In groundwater at 15 °C,

a 2 log₁₀ reduction was observed in 45 days (da Silva et al., 2016). Overall, these results provide evidence of long persistence in water².

6.5 Soil and crops

Little is known about the occurrence of astroviruses in soils or on crops. Based on their levels in human excreta and raw sewage, it is prudent to assume that the presence and persistence of astroviruses in these media would be similar to that of other human enteric viruses. Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste. In Argentina, HAsTVs were detected by molecular-based techniques in green leafy vegetables, irrigation water and sewage, which suggests that green crops can act as potential sources of HAsTVs (Prez et al., 2018).

6.6 Surfaces

On surfaces (e.g. in food preparation or health-care settings), astroviruses have been shown to survive for up to 2 months, with 4 log₁₀ and 5.3 log₁₀ reductions in the presence and absence of faecal material, respectively (Kramer et al., 2006). However, studies of astroviruses in a 20% faecal suspension at 20 °C on a toilet seat or filter paper showed a 2 log₁₀ reduction in virus concentrations in 1 day or less; at 4 °C, the same reduction took 3 days (Abad et al., 2001).

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of astroviruses (da Silva et al., 2016). However, given the importance of person-to-person and fomite transmission, additional control measures, including adequate hygiene, are needed to effectively prevent transmission (Kramer et al., 2006).

8 Significance in drinking-water

HAsTVs are commonly found in surface water and have been detected by molecular-based methods, albeit infrequently, in groundwater and treated drinking-water. In an epidemiological study of drinking-water in households in France, an increased risk of gastrointestinal illness was associated with the presence of astroviruses (Gofti-Laroche et al., 2003). However, the human health effects data from this study were limited. In an outbreak in the Republic of Korea (Hwang et al., 2015) contaminated drinking-water was identified as one of several other possible sources of contamination (food, infected food handler and cookware). While it is plausible that HAsTVs are transmitted through drinking-water, there is only weak evidence of confirmed waterborne disease outbreaks attributed to this pathogen.

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from astroviruses.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including astroviruses. Astroviruses have moderate resistance to oxidizing chemical disinfectants such as chlorine but probably have low resistance to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of astroviruses by wastewater treatment processes is variable, although the data are limited (de Silva et al., 2016). Based on available evidence, the reduction of astroviruses by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli*.

There are no data available on removal of astroviruses by onsite systems. In centralized sewage treatment systems, activated sludge treatment achieved a mean of 2.7 log₁₀ reduction of astrovirus gene counts in one study and 3 log₁₀ removal in two other studies (Le Cann et al., 2004; Aw & Gin, 2010; Fumian et al., 2013). More advanced treatment involving primary sedimentation, activated sludge and chlorination achieved a mean 2.5 log₁₀ reduction of infectivity and 3.8 log₁₀ reduction in gene counts (Morsy El-Senousy et al., 2007), while activated sludge, chlorination and sand filtration achieved a mean 2.4 log₁₀ reduction in gene counts (Hata et al., 2015).

9.2 Drinking-water supplies

Although the evidence that HAsTVs are transmitted through drinking-water is weak, water safety plans (WHO, 2012; 2023) that include control measures to protect source waters from human faecal wastes, apply adequate water treatment processes and protect drinking-water quality during distribution and storage will reduce their potential presence in drinking-water.

It is likely that removal of HAsTVs by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration) will be similar to that reported for other enteric viruses (Teunis et al., 2009). Moreover, conventional treatment can remove 1.5 log₁₀ of noroviruses (Shin & Sobsey, 2015), which are also non-enveloped RNA viruses of similar size as HAsTVs. Disinfection with initial free chlorine doses of 0.5 mg/L and 1 mg/L reduced astrovirus concentrations in water by 2.5 log₁₀ and 3 log₁₀ units, respectively in 1 hour at 20 °C and pH 7.5 (Abad et al., 1997). Based on this limited information Ct (product of disinfectant concentration and contact time) values are on the order of 30 mg.min/L. Accordingly,

astroviruses have a moderate resistance to chlorine,³ although it can be effective provided adequate doses and contact times are applied. The sensitivity of astroviruses to UV light disinfection is not known, but 2 log₁₀ inactivation of other enteric RNA viruses was achieved at doses ranging from 11 mJ/cm² to 21 mJ/cm² (Hijnen et al., 2006). Therefore, astroviruses are probably sensitive to UV light.

Due to the smaller size of HAsVs relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their longer environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAsVs in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAsVs.

10 ICD-11 classification

1A21	Gastroenteritis due to Astrovirus
1A2Y	Other specified viral intestinal infections due to Astrovirus

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.